

Shreyas Gowda Marigowda

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Summary

Automation and robotics engineer with an M.Sc. in Robotics and Automation, a B.E. in Mechatronics and 3+ years of work experience delivering industrial simulation, virtual commissioning, and automation projects for clients across Sweden and the USA. Experienced end-to-end: virtual design, PLC and robot programming, structured test campaigns, on-site commissioning and FAT. Currently working as an Automation Consultant supporting live battery assembly production at Volvo Cars Torslanda — troubleshooting robots, PLCs, conveyors, vision, gluing, welding and laser systems, and developing SOPs, checklists and supporting root cause analysis for the design team.

Experience

- May 2026 – Present** **Automation Consultant – Cleeven SC AB [Client – Volvo Cars]**
Torslanda, Gothenburg
- **Production Support & Troubleshooting:** Provide hands-on technical support for live battery assembly, resolving faults across robots, PLCs, conveyors, automation systems, and mechanical/pneumatic/hydraulic infrastructure to minimize downtime.
 - **Control Chain Diagnostics:** Diagnose issues from field-level devices and sensors through PLC logic and robot programs to HMI sequencing, focusing on root-cause analysis rather than temporary fixes.
 - **Specialized Process Maintenance:** Maintain and troubleshoot advanced process equipment, including gluing, welding, laser systems, and machine vision applications for quality and position verification
 - **Documentation & Cross-Functional Collaboration:** Author SOPs and maintenance checklists to capture technical knowledge, while partnering with production, quality, and engineering teams in a safety-critical environment.
- Sep 2023 – May 2026** **Application and Simulation Engineer – Wiksfors Technology AB**
Värmlands Nysäter, Sweden
- Delivered production simulation projects using **Siemens Tecnomatix Plant Simulation** and **Visual Components** covering DES, capacity analysis, buffer sizing, cycle time optimization, bottleneck identification, line balancing, OEE analysis, and virtual commissioning for clients in Sweden and USA
 - Commissioned robot cells and PLC-controlled machines end-to-end: virtual design → PLC logic → test campaign → on-site FAT, including on-site installation, calibration and customer hand-over
 - Programmed and virtually commissioned control logic in **Beckhoff TwinCAT 3** (ST, LD, FBD, SFC) and Mitsubishi GX Works 3, wired to simulation models over **OPC UA**
 - Built production-ready desktop applications in Visual Studio using **C#, WPF** and **XAML** for manufacturing automation workflows
 - Acted as technical bridge between customer requirements and engineering delivery across multiple simultaneous projects
- Jan 2023 – Jun 2023** **Master Thesis – Stora Enso AB**
Grums, Sweden
- Title: **Discrete Event Simulation-based Optimization of Sawmill Production** [Link](#)
- Created a discrete event simulation (DES) model of the sawmill plant using Tecnomatix Plant Simulation
 - Experimented with the model and analyzed different scenarios to identify areas for improvement
 - Investigated how the results of the simulation can be utilized to inform decision-making in the sawmill, validating the virtual model against real production data
- Sep 2021 – Nov 2021** **Internship Project - Click-In (Fixture Systems AB)**
- Title: Automatic Window mounting with the Click-In system [Link](#)
- Created a safe robotic cell for automatic window mounting using a virtual design and offline robot programming, which was then tested on an ABB IRB440 robot
 - Presented it during “Automation Day (2021), a Tech InWest Sweden” event, in collaboration with Click-In Fixture Systems AB at **“Production Technology Centre, Innovatum, Trollhättan, Sweden”**
- Nov 2019 – Apr 2020** **Production Engineering Intern - Chaitrika Engineering Works**
Bengaluru, Karnataka, India
- Hands-on experience in Lathe and CNC machine operations
 - Performed Cutting, Drilling & Turning Operations and worked on CNC Machines (**Jyoti DX200 3B & AM Spinner 15 DX**)
- June 2015 – May 2016** **Internship Trainee - Yuken India Limited**
Bengaluru, Karnataka, India
- **Quality Control & Metrology:** Served as Quality Control Inspection Engineer for valves and motors, applying **Six Sigma** concepts, calibration and high-precision testing using **ZEISS Contura G2** (CMM), Parker icountLaserCM20 and Mitutoyo SJ-400 & RA2000 surface and roundness testers
 - **Production Development:** Analysed material factors, design tolerances and production schedules to streamline the development of hydraulic components
 - **CNC Operation & Machining:** Full machine setups, tooling changes and machining sequences on Fanuc, BFW and AGNI CNC machines
 - **Systems Assembly:** Precision assembly of complete Hydraulic Powerpacks, integrating pumps, motors, gauges, accumulators and hydraulic valves

Skills & Software

- **PLC & HMI:** Beckhoff TwinCAT 3, Siemens TIA Portal, Mitsubishi GX Works 3
- **Communication:** OPC UA, Profibus/Profinet, TCP/IP
- **Robotics:** ABB RobotStudio (online/offline), KUKA KRL, Universal Robots (e-Series, CB3), ROS
- **Tools:** Azure DevOps, Git, draw.io
- **Simulation & Digital Twins:** Visual Components, Siemens Tecnomatix Plant Simulation, Siemens Process Simulate
- **Software:** Visual Studio (C#, .NET, XAML, WPF), Python 3, MATLAB R2021
- **CAD & 3D:** SolidWorks 2021, PTC Creo 8.0, Siemens NX, Blender, Twinmotion 2025

Work Projects [\[Wiksfors Technology\]](#)

- Sep 23 – Nov 23** **Simulation, Testing & Commissioning** [\[Verkstadsbolaget AB, Säfte, Sweden\]](#) [Link](#)
- Built a full kinematic model of the SkruvPortal machine in Visual Components and virtually commissioned PLC logic in TwinCAT 3 via OPC UA for nailing, routing, tool-zone collision prevention, and E-stop functionality
 - Co-developed .NET/WPF BTL data-handler app; ran 40+ production test cases and delivered approved FAT
- Nov 23 – Jan 24** **Roof Truss Factory Simulation** [\[House of Design Robotics, Idaho, USA\]](#) [Link](#)
- Parametric Visual Components simulation of AMD-1000 robotic assembly line: integrating ABB IRB 6700 (truss board handling) and ABB IRB 1200 (nail-plate placement)
 - Delivered cycle time, robot utilization, and buffer metrics per configuration, covering all six product variants from a single parametric model
- Jun 24 – Sep 24** **Factory Flow Simulation** [\[Zennihome, Arizona, USA\]](#) [Link](#)
- Modelled full modular wall panel production flow in Visual Components covering material flow, station utilization, buffer sizing, and throughput analysis to support line layout decisions
- Aug 24 – Sep 25** **Software Development + Simulation** [\[Sustainacon AB, Energimyndigheten, Säfte Bostäder AB\]](#) [Link](#)
- Built RenoChain: a custom 3D desktop application combining factory planning with simulation, developed in C#, WPF and XAML
 - Featured in an official Visual Components case study and covered in Säfte-Tidningen (Aug 2024)
- Oct 24 – Dec 24** **Robot Cell Design and Installation** [\[Heart Aerospace, Gothenburg, Sweden\]](#)
- Designed & installed 7-axis KUKA robot cell (KR 300 R2500 Ultra SE on KL linear track) in Visual Components, validating layout, work envelope and safety distances before build
 - On-site installation: robot base and track positioning, fence and light-curtain installation, electrical-cabinet wiring against EPLAN drawings, stop-time verification and KUKA brake-test commissioning
- Nov 24 – Sep 25** **Software Development & Support** [\[Autovol, Nampa, Idaho, USA\]](#)
- Maintained custom automation software for modular prefab housing factory workflows
 - Delivered feature updates and integration support over 10 months
- Feb 26 – Mar 26** **Wall Panels Production Simulation** [\[Finnebäcks AB, Edane, Sweden\]](#)
- Production simulation of wall panel production covering buffer sizing, station cycle times, and material feed sequencing to support capacity planning

Academic Projects [\[Högskolan Väst\]](#)

- Aug 21 – Jan 22** **Automated Part Assembly Line Automation**
- Programmed an ABB robot and TwinCAT PLC using SFC and Structured Text, implementing state machines (Auto, Manual, Pause, Emergency) with Profibus communication, and integrated MATLAB machine vision over TCP/IP for coordinate-based pick-and-place of colour-matched parts across 3 buffer locations
 - Developed colour-based pallet acceptance, assembly sequencing for multiple part types, and error-proof ladder logic with TON timers — including automated rejection handling, missing part notifications, and automatic Pause Mode on failure
 - Architected OPC-UA and Profibus communication between PLCs, ABB robots, and an AGV (MiR 200) for docking and order fulfilment, and designed an HMI for task control, status monitoring, and mode switching
- 2020 – 2022** **Academic Assignments**
- Built ROS workspaces using Python & MATLAB with SLAM for object detection and path planning, and implemented a goal-oriented Multi-Agent System (MAS) on the CMAS platform for autonomous conveyor control applying Industry 4.0 and CPPS principles
 - Gained experience in smart sensing, signal processing, dynamics, trajectory planning, and motion planning for autonomous robotic systems

Education and Training

- June 2023** **Master's Degree (MSc): Robotics and Automation: [120 credits]**
Högskolan Väst (University West), Trollhättan, Västra Götaland, Sweden
- May 2019** **Bachelor of Engineering (BE): Mechatronics Engineering: [180 credits]**
Mangalore Institute of Technology and Engineering, Mangalore, Karnataka, India
- June 2023** **SFI – Kurs C [Folkuniversitetet, Karlstad, Sweden]**

Certifications

- 2022** **Universal Robots Academy** [\[Universal Robots\]](#)
e-Series Core Track, e-Series Application Track, e-Series Pro Track, CB3 e-Learning and UR Cobot Summer School 2022
- 2016** **Industrial Automation Technology** [\[Bosch Rexroth Drive & Control Academy, MITE\]](#)
Certification programme in Hydraulics & Pneumatics, completed in line with Bosch Rexroth Training Centre guidelines
- 2017** **NX – Essentials for Designers** [\[Siemens Industry Software – Certificate of Merit\]](#)
- 2018** **Automation Workshop – PLC & SCADA** [\[SMEC Automation / SMEC labs\]](#)
PLC: ABB 07KR51 & SCADA: Schneider Electric Vijeo Citect
- 2022 - 2026** **Udemy Certifications** [\[Online\]](#)
PTC Creo Parametric – Complete Beginner to Advanced Course & Python Programming Masterclass.
Certified Lean Six Sigma Green Belt [\[Advanced Innovation Group Pro Excellence \(AIGPE\)\]](#)

Languages

- English (fluent)
- Swedish (SFI Kurs D)*
- Kannada (native)
- Hindi (proficient)